

SIH PROBLEM STATEMENT ANALYSIS

SIH prototype could have:

1. Input

- Drone video
- Camera calibration/intrinsic parameters
- GPS/IMU data if available

2. AI/computer vision

- Extract frames
- Feature detection and matching
- Camera pose estimation
- Depth estimation
- Multi-view/stereo reconstruction
- Handle motion blur and changing viewpoints

3. 3D reconstruction

- Generate point cloud
- Generate mesh
- Apply image textures
- Scale/georeference the model

4. Dashboard

- Upload drone video
- Processing progress
- 3D model viewer
- Measurements such as height, length and area
- GPS/map position
- Export .OBJ, .PLY, .GLB, etc.

STATEMENT:

Problem
Statement ID 26158

**Problem
Statement
Title**

Single-Pass Drone Video to Accurate 3D Model Generation System

- Background:

Generation of accurate 3D models of buildings, infrastructure, terrain, and objects typically requires multiple drone passes, extensive image overlap, specialized flight planning, and significant post-processing time. In operational scenarios such as disaster response, surveillance, infrastructure inspection, military reconnaissance, and rapid mapping, there is often only a single opportunity to capture data over the target area. A solution capable of generating an accurate and textured 3D model from a single drone pass video would significantly reduce mission time, operator effort, data acquisition requirements, and processing complexity while enabling near real-time situational awareness.

- Description:

Design and develop an AI-enabled system capable of generating a georeferenced and metrically accurate 3D model of a scene using only a single-pass drone video stream captured from a moving UAV. The system should process video frames captured during one flight path and reconstruct:

(i) 3D terrain and structures (ii) Building facades and rooftops (iii) Roads and infrastructure (iv) Vegetation and obstacles (v) Textured 3D meshes or point clouds

Description

- Expected Solution/Deliverables:

The generated model should be suitable for visualization, measurement, and analysis purposes.

- Key Challenges (i) Limited viewing angles due to single flight path.

(ii) Motion blur and video compression artifacts.

(iii) Variable illumination and shadows.

(iv) Dynamic objects (vehicles, humans, animals).

(v) GPS inaccuracies and sensor noise.

(vi) Real-time or near-real-time processing requirements.

(vii) Reconstruction of occluded surfaces.

(viii) Maintaining metric accuracy without extensive Ground Control Points (GCPs).

- Input Data :

- Mandatory (i) Drone video (1080p/4K)

(ii) GPS coordinates (iii) Flight metadata

- Optional (i) IMU data (ii) Barometric altitude (iii) Camera intrinsic parameters (iv) RTK/PPK corrections Add 'Desired Output' and 'Evaluation Criteria' table here

- Potential Applications :

(i) Border and strategic area mapping (ii) Disaster damage assessment (iii) Urban planning and smart cities (iv) Infrastructure inspection (v) Construction progress monitoring (vi) Archaeological documentation (vii) Digital twin generation (viii) Military reconnaissance and mission planning

Organization National Technical Research Organisation (NTRO)

Department National Technical Research Organisation (NTRO)

Category Software

Theme Robotics and Drones

Youtube Link

Dataset Link Will be provided real time.

I'm reading "viewpoint technique" as **Next-Best-View (NBV) planning** — active, real-time viewpoint selection where the drone doesn't fly a rigid pre-planned path, but instead steers itself (or its gimbal) toward whatever angle will reduce the biggest gap in the reconstruction *as it's building*, live, mid-flight. Let me know if you meant something else, but this fits the conversation well, so here's the case for it.

Does this actually work, and does it break "single pass"?

It works, and it doesn't break single-pass — that's the key nuance. NBV research on UAV 3D reconstruction specifically shows a prediction-based NBV approach reaching 88% surface coverage compared to ground truth and a 37.5% efficiency gain over traditional pre-planned flight paths, using real-time trajectory adjustment based on predictions of what's still unseen. That's not "fly it twice" — it's "make the *one* flight smarter about where it looks," which is fully compatible with the single-pass constraint in the brief. In fact, it directly attacks the root cause of limited angles rather than working around it after the fact.

How it fits with what you've already got

Think of it as sitting **upstream** of the three techniques you listed — it changes what gets captured in the first place, so there's less for symmetry-completion and diffusion fallback to even have to fix:

1. **NBV / active viewpoint planning** (new, upstream): live reconstruction confidence map → drone/gimbal steers toward the weakest-coverage regions in real time during the single flight.

2. **Flight-path optimization** (oblique + weave): your static baseline plan, running by default even before NBV kicks in.
3. **Symmetry/structural priors**: fills whatever gaps remain after 1+2, cheaply and reliably.
4. **Diffusion completion (AnyCity-style)**: last-resort fallback for anything symmetry can't resolve, always tagged "inferred."

That's a genuinely defensible pipeline: minimize the gap actively during capture → minimize it further with cheap geometric reasoning → only generate when truly necessary → always disclose what's generated.

Feasibility reality-check for a SIH prototype

- **Full onboard real-time NBV (gimbal/flight-path control loop) is the ambitious version** — it needs a live confidence estimate of the reconstruction *while flying*, which is genuinely hard to get running end-to-end in a hackathon timeframe, and needs the drone's flight controller to accept live waypoint/gimbal updates.
- **A scaled-down, very achievable version**: simulate it. Since your dashboard already processes video, you can build an NBV *module* that, given the growing reconstruction, computes and visualizes "if we had steered the gimbal here next, we'd have covered this facade" — this demonstrates the concept, the math, and the value without needing real flight-control hardware integration. That's honest, buildable, and still a strong differentiator in a demo.
- **Occlusion-aware scoring is the computationally cheap part** — shadowcasting/visibility-based scoring (used in several NBV papers) is much faster to implement than a learned RL-based NBV policy, and is the right choice given your time budget.

If you're building a prototype rather than a production drone system, I'd pitch it as: "our system doesn't just process video passively — it identifies its own blind spots during flight and would actively steer to close them; here's a simulated demonstration of that logic." That's honest about what's implemented vs. designed-for, and it's a strong technical story.

SOLUTION APPROACH IDEA

Final Solution Report — SIH26158

Single-Pass Drone Video to Accurate 3D Model Generation System

1. Proposed Solution

Intelligent, Occlusion-Aware 3D Reconstruction from a Single Continuous Drone Flight

The proposed system converts a **single continuous drone flight** — one pass, no re-flights — into a **georeferenced, metrically usable 3D model**, while being explicit about which parts of that model are directly observed, which are inferred, and which are simply missing.

The system has **two equally important halves**, and the first version of this report only covered one of them:

1. **Capture-efficiency pipeline** — intelligently select which frames from a continuous video stream actually contribute new spatial information, and establish reliable correspondence between them.
2. **Occlusion/limited-viewpoint pipeline** — actively minimize, and then honestly account for, the fact that a single pass physically cannot see every angle of every structure.

Presenting only the first half (as the earlier version of this report did) leaves an obvious gap: intelligent frame selection makes reconstruction *efficient*, but it does nothing about the *fundamental single-pass constraint* — limited viewing angles — which is the central challenge named in the problem statement itself. This version corrects that.

The resulting model supports: strategic/border-area mapping, disaster damage assessment, infrastructure inspection, urban planning, construction monitoring, archaeological documentation, digital-twin generation, and reconnaissance/mission planning.

2. Problem

A drone captures a continuous video stream during one flight. Two distinct classes of problems arise from this, and they need different solutions:

2A. Data-volume problems (solved by the capture-efficiency pipeline)

- Many frames are redundant.
- Motion blur makes some frames unusable.
- Consecutive frames may contain very little new information.
- Reconstruction becomes computationally expensive as frame count grows.
- Dynamic objects (vehicles, people) introduce incorrect feature matches.

2B. Coverage problems (solved by the occlusion-handling pipeline — under-addressed previously)

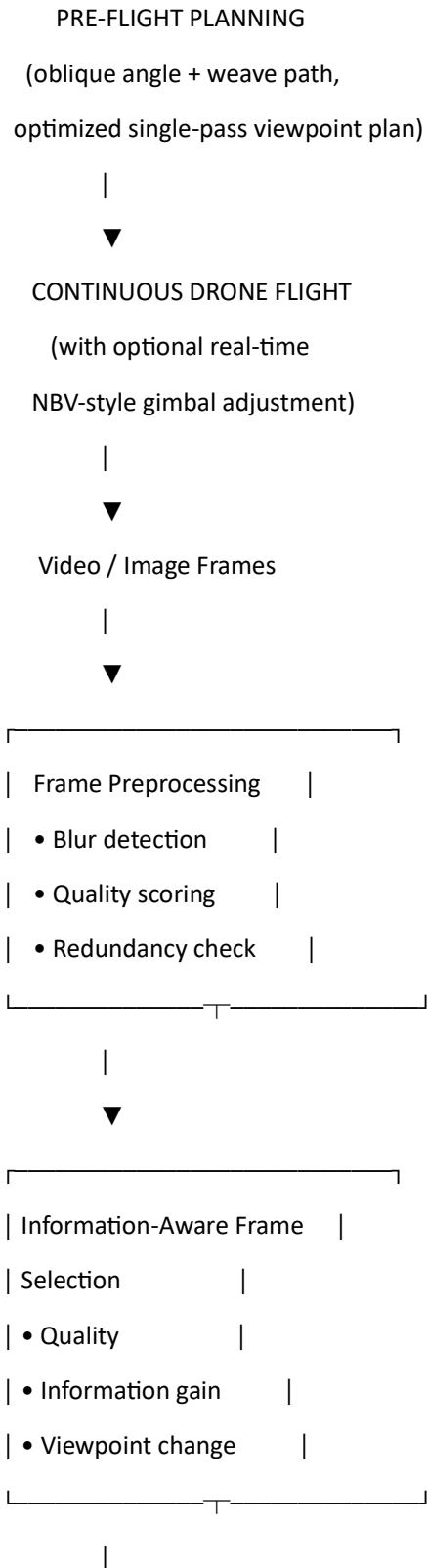
- A single pass gives limited viewing angles by design — there's no second flight to fill gaps.
- Building facades, undersides, and occluded surfaces may never enter the camera's field of view at all.
- No amount of clever frame *selection* recovers geometry that was never *captured* in the first place.

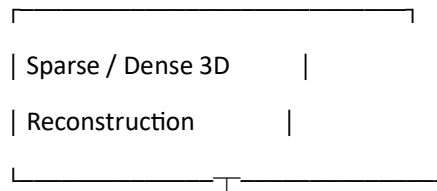
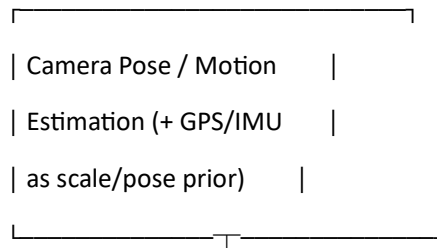
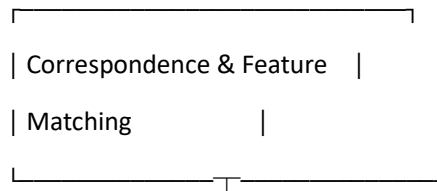
The real challenge, stated precisely, is therefore two questions, not one:

(1) How do we intelligently select and correspond useful observations from a continuous single-pass flight? (2) Given that a single pass inherently limits viewing angles, how do we minimize coverage gaps during capture, and transparently handle whatever gaps remain?

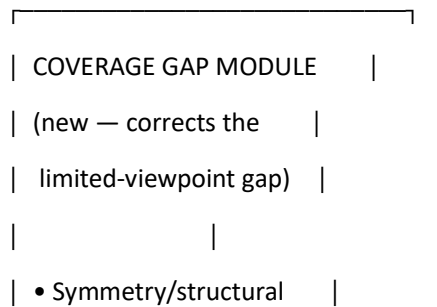
Answering only (1) — as commercial SfM/keyframe pipelines already do — is not a complete solution to PS26158. Answering (2) is where the problem statement's actual difficulty, and your genuine novelty, lives.

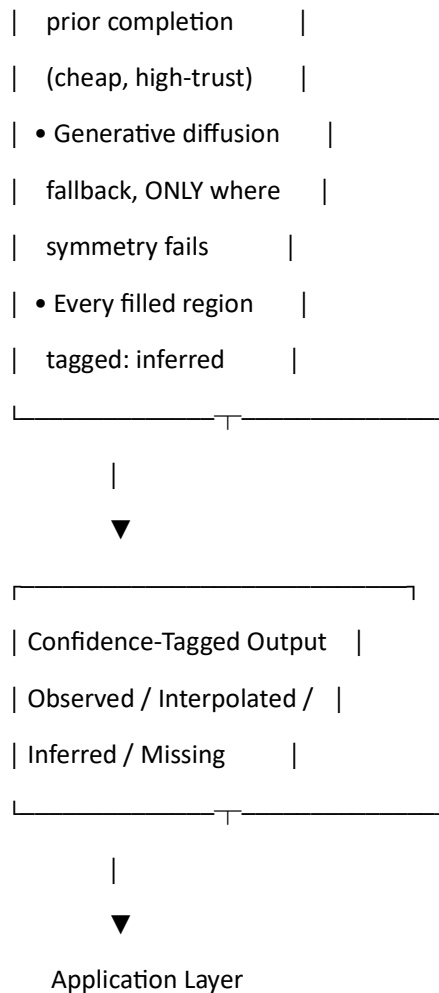
3. Proposed System Architecture (Revised)





3D POINT CLOUD / MESH
(with per-region coverage
confidence map)





4. Core Innovation

This project should **not** be presented as "we use OpenCV/COLMAP + a drone" — that's integration of existing tools, not a contribution. It should also **not** be presented as only "smart frame selection" — that solves efficiency but leaves the problem statement's headline challenge (limited viewing angles) unaddressed.

Corrected novelty statement

An information-aware, occlusion-conscious reconstruction pipeline that (a) selectively processes continuous single-pass drone observations based on quality, spatial novelty, and correspondence potential, and (b) actively minimizes and then transparently accounts for the coverage gaps that a single flight path inherently creates — through pre-flight path optimization, in-flight adaptive viewpoint steering, and confidence-tagged gap-filling — rather than presenting a silently completed model.

This gives you two connected, defensible novelty claims instead of one incomplete one:

Claim	What it solves	Why it's defensible
Information-aware frame selection	Data volume / redundancy / compute cost	Tuned specifically for single-pass UAV footage (motion blur, GPS-timestamp-aware overlap), not a generic keyframe filter
Occlusion-aware coverage pipeline	Limited viewing angles (the problem statement's core challenge)	Directly answers what the earlier version of this report left as an open weakness

5. Frame Selection Module (Data-Volume Half)

Given a continuous stream $F_1 F_2 F_3 \dots F_n$, most frames are near-duplicates. The system scores each frame and keeps only those that add real value:

Frame Utility = $w_1 \cdot (\text{Image Quality}) + w_2 \cdot (\text{Feature Richness}) + w_3 \cdot (\text{Viewpoint Change}) + w_4 \cdot (\text{Information Gain}) + w_5 \cdot (\text{Matchability})$

To make this a real contribution rather than a generic formula, the weights and signals must be tuned for the single-pass UAV case specifically:

- **Image Quality:** blur score should be weighted more heavily than in ground-based SfM, because UAV vibration and forward motion at typical survey speeds (5–10 m/s) causes far more motion blur per frame than a handheld or tripod capture.
- **Viewpoint Change:** computed using GPS/IMU-timestamp deltas as a cheap prior *before* doing expensive feature matching — only run full feature extraction on frames that GPS suggests moved enough to matter.
- **Information Gain:** penalize frames dominated by texture-poor regions (sky, water, uniform road surface, in your own project description's use cases) since these contribute little to feature-based reconstruction regardless of quality.
- **Matchability:** frames near the edges of the flight path (turns, altitude changes) get a matchability boost check, since these are disproportionately useful for filling angular gaps — this is the direct link between the frame-selection module and the coverage module below.

Frames below a threshold are discarded; frames that specifically improve angular coverage of a weakly-observed region are protected from discarding even if otherwise redundant.

6. Coverage-Gap Module (Limited-Viewpoint Half — Corrected Focus)

This is the module the earlier report was missing. It runs in three phases, matched to when in the mission each technique is possible:

6A. Pre-flight: optimized single-path planning

- Fly with an **oblique camera angle** (not purely nadir/straight-down) — oblique passes capture vertical surfaces (building facades, infrastructure sides) that a straight-down pass never sees.

- Use a **zigzag/weave lateral deviation** within the single pass rather than a perfectly straight line, to gain stereo baseline diversity for free.
- Where site geometry is roughly known in advance (from GPS coordinates + any available coarse map data), pre-compute a single optimized path — sequencing viewpoints to maximize coverage and triangulation quality within one continuous flight, rather than flying a naive straight line.
- This is cheap, requires no exotic AI, and should be the default in every mission — it reduces how much the later modules ever need to fix.

6B. In-flight: adaptive viewpoint steering (stretch goal for prototype)

- As reconstruction confidence is estimated live during flight, steer the gimbal (or, if hardware allows, the flight path itself) toward the currently weakest-coverage regions — a Next-Best-View-style adjustment, but applied *within* the single continuous flight, not as a second pass.
- For a hackathon prototype, this can be **simulated and visualized** rather than hardware-integrated: show, after the fact, where the system would have steered next and why — this demonstrates the concept honestly without requiring live flight-controller integration.

6C. Post-flight: transparent gap-filling

Whatever coverage gaps survive phases A and B are filled using a strict priority order, each tagged in the final output:

1. **Interpolation** between nearby observed points — for small gaps, minimal risk.
2. **Symmetry/structural priors** — most man-made structures (buildings, roads, infrastructure) have exploitable regularities: bilateral symmetry, repeated floors/windows, planar rooftops. Detecting a symmetry axis from the observed side and mirroring it onto an occluded side is cheap, explainable, and far more trustworthy than a generative model for structured objects specifically.
3. **Generative diffusion completion** — used *only* when symmetry/structural priors cannot resolve the gap (irregular terrain, vegetation, damaged structures with no intact mirror side). Every pixel/point filled this way is explicitly a plausible estimate, not a measurement.

6D. The non-negotiable output rule

Every point/face in the final model carries one of four tags:

- **Observed** — directly triangulated from real imagery.
- **Interpolated** — filled between nearby observed points.
- **Inferred (symmetry)** — completed using a structural/symmetry prior.
- **Inferred (generative)** — completed using a diffusion/generative fallback.

This is displayed as a color-coded confidence overlay in the dashboard. This single feature is what turns "the AI might be hallucinating a wall that isn't there" from an unanswered criticism into a design decision you can defend on stage.

7. Image Correspondence

After frame selection, the system matches image regions representing the same physical location using standard feature techniques (SIFT, ORB, SuperPoint, LightGlue, LoFTR). These are **components**; the contribution is how they're integrated with the frame-selection scores and coverage-confidence tracking from Sections 5–6 — for example, using matchability as direct input to the frame-utility score, and feeding correspondence density directly into the per-region confidence map that drives the coverage-gap module.

8. Camera Motion Estimation

Camera pose is estimated from matched features; GPS/RTK/IMU, if available, provide additional priors for:

- Georeferencing
- Scale (critical since monocular/visual reconstruction is scale-ambiguous without it)
- Camera pose priors
- Alignment

GPS/RTK should not be presented as performing the reconstruction itself — it primarily anchors scale and location. The imagery provides the actual scene geometry. (See our earlier discussion: base-map/satellite data can supplement this further, but only as an optional, connectivity-dependent enrichment layer — never a dependency for core operation, since field connectivity can't be assumed for border/disaster scenarios.)

9. 3D Reconstruction Pipeline

2D Features → Correspondence → Camera Pose → Triangulation

→ Sparse Point Cloud → Dense Reconstruction → 3D Mesh

→ Coverage-Gap Module (Section 6) → Confidence-Tagged Output

Output formats: sparse point cloud, dense point cloud, mesh, textured model — each carrying the confidence tags from Section 6D.

10. Handling Difficult Frames (Data-Volume Side)

Condition	System behavior
Blur	Discard / downweight
Low information (sky, water, uniform surface)	Lower reconstruction utility
Redundant frames	Removed, unless they uniquely improve angular coverage (see Section 5)

Condition	System behavior
Poor overlap	Not forced into reconstruction
Dynamic objects	Identified/masked, matches downweighted

11. The Fundamental Limitation, Stated Precisely

3D reconstruction can recover geometry from observed visual information. It cannot guarantee accurate reconstruction of genuinely unobserved regions — no matter how good the frame selection is.

This is why Section 6 exists as a first-class module rather than an afterthought. The corrected position is:

- **Observed** → reconstructed directly.
 - **Weakly observed** → interpolated, low uncertainty.
 - **Occluded but structurally regular** → completed via symmetry prior, tagged as inferred.
 - **Occluded and irregular** → completed via generative fallback *only if attempted*, tagged as inferred, with the option to leave it explicitly marked "missing" instead if the mission calls for zero speculative content (e.g., a strict measurement/legal-evidence use case).
-

12. On "Plausible View" Generation

Generating a plausible image from an unobserved viewpoint (via novel-view synthesis / diffusion) is **not** the same as discovering true hidden geometry. Always separate, in both the system output and the presentation:

- **Reconstructed geometry** — supported by actual observations.
 - **AI-inferred/synthesized geometry** — estimated from learned visual/structural patterns, explicitly labeled.
-

13. Digital Twin Output

Real Environment → Drone Observation (single pass, optimized path)

→ Frame Selection & Correspondence → 3D Reconstruction

→ Coverage-Gap Resolution (tagged) → 3D Spatial Model → Digital Twin

Supports measurement, inspection, visualization, change detection, planning, and simulation — with every derived measurement able to state whether it's based on observed or inferred geometry.

14. Application Mapping

Border & strategic area mapping — terrain understanding, elevation analysis, route/obstacle planning. Keep framing at mapping/visualization/planning level, not operational decision-making.

Disaster damage assessment — compare before/after 3D models for change detection; this is a strong showcase for the confidence-tagging system, since distinguishing "genuinely collapsed" from "occluded and unknown" matters operationally.

Urban planning — building visualization, road/spatial/infrastructure planning.

Infrastructure inspection — bridges, towers, buildings, roads, industrial structures.

Construction monitoring — repeated flights compared over time (Day 1 → Day 15 → Day 30).

Archaeology — non-contact documentation of monuments, ruins, excavation sites.

Digital twins — spatial layer for simulation and planning.

Reconnaissance & mission planning — terrain understanding, route planning, obstacle awareness; keep at mapping/visualization level for the presentation.

15. Hardware–Software Architecture

Input: Drone camera / recorded drone flight

|

Pre-flight: Optimized single-path plan (oblique + weave)

|

Processing: Laptop / workstation

|

AI/CV: Frame quality analysis → Frame selection

→ Feature extraction/correspondence → Pose estimation

→ Reconstruction → Coverage-gap resolution

|

Output: Confidence-tagged 3D point cloud / mesh / digital twin

Optional metadata: GPS, RTK GNSS, IMU, altitude, timestamp.

16. What Happens if Required Footage Was Never Captured?

"The software system operates on the visual data captured during the single drone pass. Where a region was never captured — due to hardware, environmental, or operational limitations — that information is inherently unavailable to the reconstruction system. We treat data acquisition as an input constraint. Within that constraint, our system does three things competitors don't: it minimizes the chance of missing a region in the first place through optimized flight planning, it attempts a

conservative, symmetry-based recovery where structurally justified, and — critically — it never silently presents a guess as a measurement."

This is a stronger answer than the earlier version because it doesn't just disclaim the limitation — it shows three concrete mitigations before falling back to "this is a hardware/data constraint."

17. Expected Prototype (Revised Scope)

Recorded Drone Video

- Simulated flight-path optimization overlay (oblique+weave demonstration)
- Frame Extraction → Blur/Quality Analysis
- Intelligent Frame Selection → Feature Matching
- Camera Pose Estimation → 3D Reconstruction
- Coverage-Gap Detection (visualize which regions are weak/occluded)
- Symmetry-Based Completion (demonstrated on at least one structure)
- Confidence-Tagged Interactive 3D Viewer

Demonstration flow

1. Original video → 2. Extracted frames → 3. Rejected frames → 4. Selected frames → 5. Feature correspondences → 6. Camera trajectory → 7. Point cloud with a visible coverage gap → 8. Symmetry-completed region, clearly tagged → 9. Final confidence-colored 3D model.

Step 7–8 is the addition that fixes the previous version's main weakness — it makes the limited-viewpoint problem, and your specific answer to it, visually undeniable to a judge.

18. Evaluation Metrics (Revised)

Reconstruction quality

Reprojection error, point-cloud completeness/accuracy, mesh quality.

Frame-selection performance

% frames removed, processing time reduction, memory reduction.

Matching

Valid match count, inlier ratio, geometric consistency.

Coverage performance (new — this was entirely missing before)

- **Angular coverage %:** fraction of surface area observed from at least one viewpoint, with vs. without flight-path optimization (Section 6A).

- **Gap-fill accuracy:** for structures where ground truth is known (e.g., a symmetric building you deliberately under-flew in testing), measure how close the symmetry-completed side is to the real geometry — this is your strongest quantitative novelty proof.
- **Confidence-tag distribution:** % of final model that is observed vs. interpolated vs. inferred — report this transparently rather than hiding it.

Comparative table

Method	Frames	Angular Coverage	Reconstruction Quality	Honest About Gaps?
All frames, naive nadir flight	High	Low	Baseline, silent gaps	No
Random frame selection	Low	Low	Reduced	No
Frame selection only (previous report)	Lower	Low	Comparable	Partial
Proposed (selection + coverage pipeline)	Lower	Higher	Comparable/better	Yes — tagged

19. Technology Stack

Programming: Python **Computer Vision:** OpenCV, feature extraction/matching libraries

Reconstruction: COLMAP / OpenMVG / OpenMVS or equivalent; optionally a feed-forward pose-free model (VGGT-family) as the base geometry estimator **AI:** SuperPoint, LightGlue, LoFTR;

segmentation/object detection for dynamic-object handling; symmetry-detection module; diffusion-based completion model as a scoped fallback **3D Visualization:** Open3D, Three.js, Unity, or Potree — with confidence-tag color overlay support **Optional geospatial:** GPS, RTK GNSS, IMU; optional connectivity-dependent base-map enrichment

20. What Makes This Different (Corrected)

Avoid claiming:

- ❌ "We use AI for 3D reconstruction."
- ❌ "We use drones for 3D mapping."
- ❌ "We use feature matching."
- ❌ "We generate a digital twin."
- ❌ "We do smart frame selection." (*on its own — this alone is now known to be an incomplete claim*)

Instead:

An information-aware, occlusion-conscious reconstruction pipeline for single-pass drone flight: it selectively processes observations to reduce redundant computation, actively minimizes coverage

gaps through optimized single-path planning and adaptive viewpoint steering, and transparently distinguishes observed from inferred geometry in every output — rather than silently presenting a complete-looking but partly guessed model.

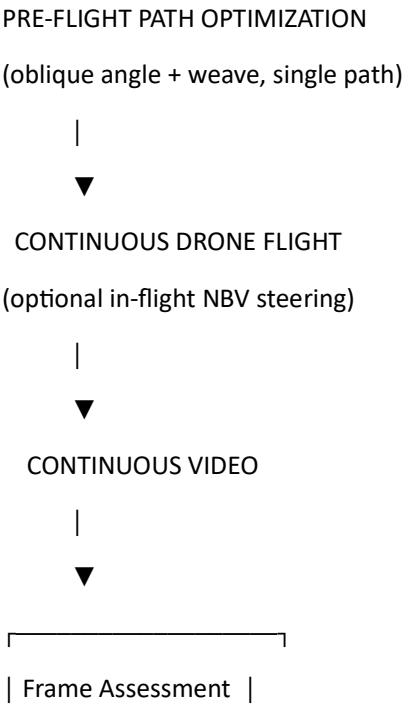
21. What You Should Not Claim

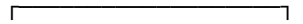
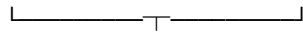
- ❌ Perfect 3D reconstruction
- ❌ Reconstruction of completely unseen environments
- ❌ No requirement for adequate drone footage
- ❌ Complete replacement of existing multi-pass photogrammetry systems
- ❌ 100% accurate reconstruction
- ❌ Full autonomous military decision-making
- ❌ That generative gap-filling recovers "true" hidden geometry (it produces a plausible estimate, always tagged as such)

Instead:

"The system produces a reliable 3D representation from available single-pass visual observations, actively minimizes coverage gaps through flight-path and viewpoint optimization, and explicitly labels any regions completed through structural or generative inference — rather than presenting a uniformly confident model that hides where it's guessing."

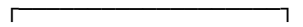
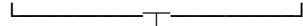
22. Final End-to-End Concept





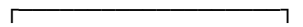
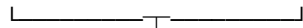
| Information-Aware |

| Frame Selection |



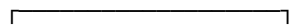
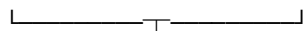
| Correspondence & |

| Feature Matching |

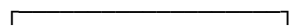
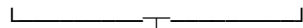


| Pose Estimation & |

| Geometric Verify |



| 3D Reconstruction |

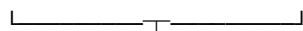


| COVERAGE-GAP |

| MODULE (symmetry |

| → generative, |

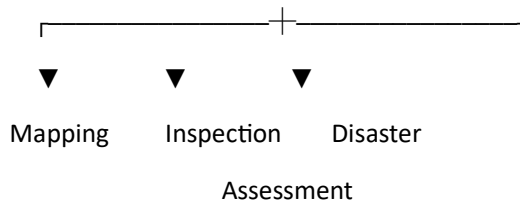
| tagged output) |



CONFIDENCE-TAGGED

3D MODEL / DIGITAL TWIN





Final one-line pitch

"We turn a single continuous drone flight into a computationally efficient, coverage-optimized, and honestly-labeled 3D model — reducing redundant frames, actively minimizing the viewing-angle gaps a single pass creates, and always distinguishing what was seen from what was inferred."

23. Assessment

Feasible, and now complete. The previous version was technically sound but incomplete — it solved the *efficiency* problem well but left the *limited-viewpoint* problem (the challenge most explicitly named in PS26158 itself) largely unaddressed. This version corrects that by making the coverage-gap module (Section 6) a first-class, equally-weighted half of the architecture, with its own dedicated evaluation metrics (Section 18) and its own line in the demonstration flow (Section 17, steps 7–8).

For SIH, build **one controlled prototype** demonstrating both halves together — frame selection *and* a visible, symmetry-completed coverage gap with confidence tagging — rather than eight separate application modules. That combination is what makes the novelty claim in Section 20 something you can actually show, not just say.